# Box by Box: Auditing a Xenogears Retranslation Against the Japanese

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**Base text: Perfect Works Build 0.11.4**

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## Abstract

Fan retranslations of Japanese role-playing games aim to bring a classic script closer to its original, yet they are usually checked against whatever source text happens to be public, and the text itself sits inside compressed binary containers that make every revision a risk. We report the statistics of **Box by Box**, a revision audit of the community Perfect Works retranslation of *Xenogears*, in which the Japanese original was decoded from the game discs and aligned with the English script one text box at a time. Revisions were proposed by SI (Synthesized Intelligence) reviewers under a conservative policy and were accepted only after a mechanical gate and a reading against the Japanese line. The audit covered 548,421 Japanese characters and 219,330 English words in 15,987 text boxes across 595 field files, with 99.7% of Japanese characters decoded. Of 502 proposals, 492 were accepted; with six consistency edits this gives 498 revisions, of which 70% correct meaning errors rather than surface errors. Revisions are small (median 22 characters) and occur at 2.27 per 1,000 English words, ranging from 1.07 to 3.04 across the game. A cross-disc comparison found 376 boxes in 20 disc-2 files that still carried the 1998 script. The results show that a full source-grounded check of a long game script is feasible without rebuilding it, and that build consistency is a translation-quality issue in its own right.


## 1. Introduction

Video-game scripts are translated under constraints that most other texts never meet: fixed text-box sizes, control codes embedded in the dialogue, and a schedule that ends with the master disc [1]. Long role-playing games are where these constraints bite hardest. *Xenogears*, released in Japan in February 1998 and in North America in October 1998 [2], carries a dense philosophical script whose 1998 English localization was produced quickly and has been debated ever since.

Fan communities answer such cases with retranslations. The Perfect Works Build [3] is a mature example: it ships a revised English script for both discs of the game, governed by a written style guide and a list of deliberate name and term choices [4]. Like any translation, it benefits from revision by a second linguist working against the source, the step that service standards make mandatory [5]. Revision is costly, though, and for a game script it is costlier still, because the source text is not a document one can open.

Two things make a full revision possible here. First, part of the Japanese script is publicly transcribed [6], which is enough to learn how the Japanese disc encodes its characters. Second, the game stores dialogue as numbered text boxes, so once both discs can be read, the English and Japanese of each box can be paired by position.

**A retranslation checked against a partial source is not checked.** The public Japanese transcript stops at chapter 6 of a game with more than sixty, so most of the script had never been compared with the original line by line.

**Fixing the text in isolation does not fix the game.** Each revision has to survive re-encoding, compression and disc rebuilding without disturbing the control codes around it, and it has to reach both discs, which number their files differently.

To address these challenges, we ran a box-by-box audit of the complete field script and report what it found. We ask four questions: how much of the script could be aligned with its Japanese source; how many proposed revisions survived the acceptance gates; what kinds of errors the revisions correct; and whether the shipped discs agree with each other. Our contributions are threefold:

- **A source-grounded corpus.** We decode the Japanese discs and align 15,987 text boxes with the English script, covering 548,421 Japanese characters. This turns a partial check into a census.
- **Revision statistics.** We report 498 accepted revisions by category, size and position in the game, together with the proposals that were rejected and why. The numbers describe where a mature retranslation still departs from its source.
- **A build-consistency finding.** We show that 20 field files on disc 2 still shipped the 1998 text for scenes revised on disc 1, and we describe the rule used to repair them. This places build checks inside translation quality assurance.

## 2. Audit design

Figure 1 gives an overview of the audit, from the four disc images to the rebuilt discs. Each stage either reads data or changes it under a gate, and nothing reaches the discs without passing every gate before it.

### 2.1. Problem formulation

We treat the audit as a tuple $\mathcal{A} = (B, J, P, G, W)$, where $B$ is the set of English text boxes, $J$ the Japanese boxes paired with them, $P$ the proposed revisions, $G = (g_1, g_2)$

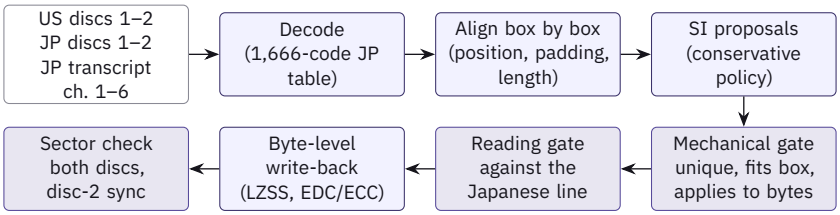



**Figure 1 | The audit reads four discs and changes text only through three gates.** Upper row: data preparation and proposal. Lower row, right to left: the mechanical gate, the reading gate and the write-back, followed by verification at sector level on both discs. Shaded boxes are gates that can reject a change. Schematic; box sizes carry no meaning.

**Table 1 | Only English box text and box size change; every other object is carried forward unchanged.**

| Object | Change rule | Carried forward |
|---|---|---|
| Japanese boxes $J$ | read only | yes |
| English box text $b$ | replace substring $o \to n$ | yes, as revised |
| Box size $(w, h)$ | may grow to fit $n$, never shrinks | yes |
| Control codes in $b$ | preserved byte for byte | yes |
| Other file sections | unchanged, except disc-2 sync (§2.4) | yes |

the mechanical and reading gates, and $W$ the write-back that turns accepted revisions into disc sectors. A proposal $p \in P$ names a box $b$, an exact substring $o$ of $b$, and its replacement $n$. It is accepted when

$$g_1(p) \wedge g_2(p) = \text{true}, \tag{1}$$

and only accepted proposals enter $W$. Table 1 lists what may change during an audit run.

## 2.2. Decoding and alignment

The Japanese disc stores most kana as single bytes and kanji as two-byte codes. We learned the code table by pairing disc boxes from chapters 1–6 with the public transcript [6] and voting on character assignments, then extended it by inferring the remaining kanji from context. The final table has 1,666 codes. Main-character names are stored as name pointers and were expanded to the names they display.

Boxes were paired by position within each field file. Eleven files did not pair at first. Eight of them turned out to end in empty boxes in the US build (29 in seven of them), and one Japanese file ends in 29 debug-menu boxes that the US build removed; after these tails were dropped, the nine files paired one to one. Two files differ by a single box, where the retranslation splits or repeats a line, and were aligned by a gapped dynamic program on box length.

## 2.3. Proposals and gates

Reviewers were given the paired boxes, the project style guide [4] and a policy: propose a change only for a mistranslation, an omission of meaningful content, a wrong or missing speaker, or a clear typo or style-guide violation, and never for tone or a more

literal rendering of a line whose meaning is already right. The project's deliberate name and term choices were protected.

**Mechanical gate $g_1$.** A proposal passes when its old text occurs exactly once in the box, every edited line stays within 28 visible characters (or the page's existing longest line), each page stays within four lines, and the encoded old text is found exactly once in the shipped bytes.

**Reading gate $g_2$.** Each surviving proposal was read against its Japanese line and accepted, adjusted or rejected. Fixes were copied to byte-identical boxes in other files only when the text was at least 40 characters long and the edit was not a speaker change; short boxes such as an ellipsis recur with different speakers.

### 2.4. Write-back and cross-disc consistency

Revisions replace the encoded substring inside the original box bytes, so control codes that no text dump exports are kept. The text section is recompressed, later section offsets are shifted, and a file that outgrows its sectors is moved to the end of the image. Sector headers, error-detection and error-correction codes are regenerated. Disc 2 uses different file indices, so files were matched by content.

A disc-2 file was replaced with its final disc-1 counterpart when three conditions held: the two files have the same number of boxes and share at least one box; the Japanese originals of the two files are byte-identical across the two Japanese discs; and the English text differs. The first condition rules out lookalike files, and the second shows that the original game uses one text for both discs.

## 3. Results

### 3.1. More than half a million Japanese characters were aligned to the English script

We compared the size of the aligned corpus with the size of the whole English script, and counted Japanese characters that the code table could not decode.

The aligned corpus covers 595 field files, 15,987 text boxes, 548,421 Japanese characters and 219,330 English words (Table 2). This is 86.6% of the 253,353 English words in the full script dump, which also contains battle and arena text outside the field files. Only 1,537 Japanese characters (0.28%) remained undecoded. Figure 2a,b splits the corpus by pass: the first pass, limited to chapters 1–6, accounts for 4.2% of the aligned English words.

The ratio of 2.50 Japanese characters per English word is stable across passes (2.48, 2.50 and 2.53), which suggests that the alignment did not systematically pair long boxes with short ones.

Three short files, two elevator menus and one Yes/No prompt, have empty text sections on both Japanese discs; their English was added in the US build and has no source to

**Table 2 | The aligned corpus covers most of the script; the mechanical pass covers all of it.** Counts are full censuses, not samples. English words exclude control codes.

| Scope | Files | Boxes | JP characters | EN words |
|---|---|---|---|---|
| Pass 1, chapters 1–6 | 18 | 593 | 22,856 | 9,209 |
| Pass 2, chapter 7 to the end | 566 | 15,047 | 509,617 | 203,808 |
| Pass 3, re-aligned files | 11 | 347 | 15,948 | 6,313 |
| **Japanese-grounded total** | **595** | **15,987** | **548,421** | **219,330** |
| Mechanical pass, whole script dump | 661 | 19,110 | — | 253,353 |
| Menu strings (disc 1) | — | 1,950 | — | 3,879 |

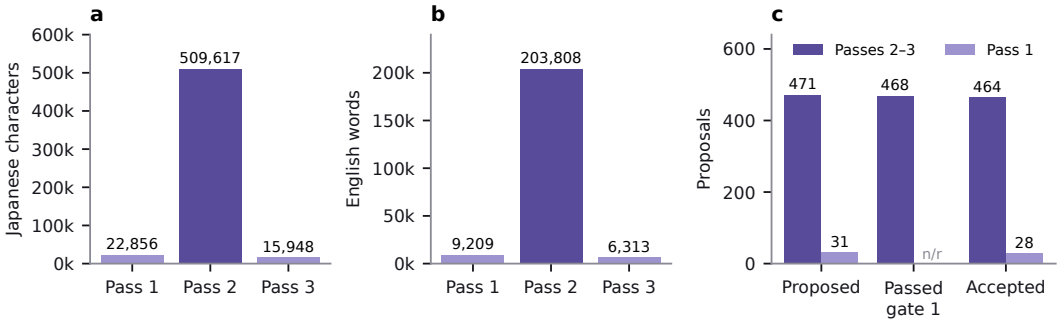



**Figure 2 | Almost every proposal passed the mechanical gate, and the reading gate removed a further handful.** **a**, Japanese characters aligned in each pass. **b**, English words aligned in each pass. **c**, Proposals at each stage. Passes 2–3: 471 proposed, 468 passed the mechanical gate, 464 accepted. Pass 1: 31 proposed and 28 accepted; no separate mechanical-gate count was recorded (n/r). Measured data.

check. One further file holds only padding.

## 3.2. The gates rejected few proposals but caught pipeline faults

We counted proposals at each stage and recorded every rejection.

Of 502 proposals, 492 were accepted (98.0%). In passes 2–3, three proposals failed the mechanical gate (0.6%) because their old text does not occur in the shipped bytes, and four more were rejected on reading (0.9%). Six revisions did not come from proposals: five aligned a champion count across files, and one was a copy of an accepted fix to an identical box in another file. Together these give 498 accepted revisions in 215 of the 595 files (36.1%).

The gates also caught three faults in the pipeline itself. A first version of the copy-to-identical-boxes rule spread a one-off speaker fix to 35 unrelated ellipsis boxes, and copied the fix for a duplicated box back onto the box that legitimately holds that line; the 40-character and other-file rules came from this failure. One proposal would have added a speaker name that an earlier pass had already added. And one box had been reflowed in the script dump but not on the disc, which the byte-level test detected.

A high acceptance rate is not evidence of high reviewer precision on its own. The policy asked reviewers to stay conservative, and the reading gate was applied by the same SI-assisted process that set up the audit, not by an independent panel.

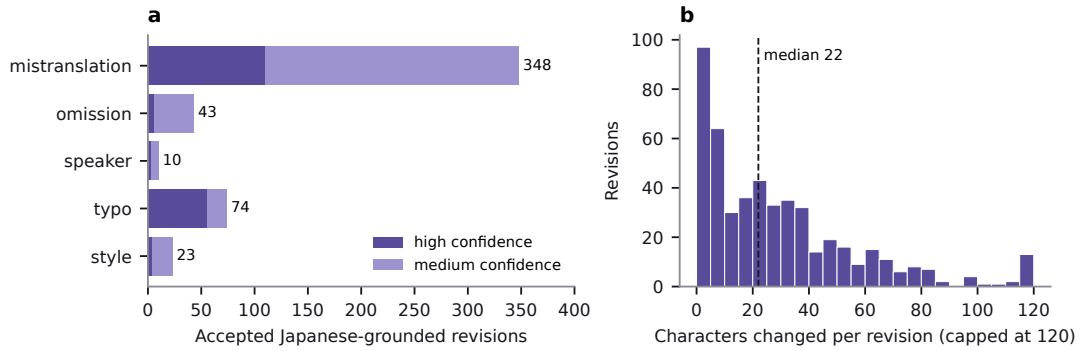



**Figure 3 | Seven in ten revisions correct meaning, and most revisions change only a few words.** **a**, Accepted Japanese-grounded revisions by category, split by reviewer confidence (n = 498). **b**, Characters changed per revision, measured as the total length of non-matching spans between the old and new text, with tags removed; values above 120 are shown at 120. The dashed line marks the median. Measured data.

## 3.3. Meaning errors, not surface errors, dominate the revisions

We classified each accepted Japanese-grounded revision by its error type and by the reviewer's confidence.

Mistranslations make up 348 of the 498 revisions (69.9%), followed by typos (74, 14.9%), omissions of meaningful content (43, 8.6%), style fixes (23, 4.6%) and speaker fixes (10, 2.0%), as shown in Figure 3a. In total, 179 revisions (35.9%) were marked high confidence. The separate mechanical pass over the whole script found surface problems instead: 75 fixes (33 typos, 42 style), 30 text boxes widened to fit their text, and 18 menu-string rules that changed 69 strings on each disc.

The meaning errors include polarity reversals and lost plot facts. Table 3 shows examples: a farewell that says the opposite of the Japanese, a radio band named as high frequency where the Japanese says long wavelength, a box that repeats another box's line instead of its own, and a dropped clause that explains how a character escaped.

These categories were assigned by the proposing reviewer and were not adjudicated against a standard typology such as MQM [7]. They map onto its Accuracy branch (mistranslation, omission) and Fluency branch (typo, style), but the counts are not MQM scores.

## 3.4. Revisions are small and local

We measured how much text each revision changed.

The median revision changes 22 characters; 48.2% change 20 characters or fewer, and 13.5% change more than 60 (Figure 3b). Across all 498 revisions the replaced spans grew from 3,769 to 3,887 words (+3.1%). Mistranslation and omission fixes have medians of 29 and 30 characters, against 1.5 for typos and 2 for style fixes.

Small edits matter for a disc game: a revision that fits the existing box needs no layout change. Only boxes whose new text needs it grow, and no box shrinks.

**Table 3 | Representative revisions.** Japanese lines are glossed in English; file and box numbers refer to the field script.

| Box | Kind | Before | After | Japanese, glossed |
|---|---|---|---|---|
| 1640#5 | mistranslation | It could do with some improvement. | It wasn't half bad. | the brief dream we spoke of was not bad |
| 1186#121 | mistranslation | We can use high frequency … | We can use low frequency … | long-wavelength radio works under water |
| 1240#8 | mistranslation | with Shakhan being so quiet … | Shakhan will not simply stay quiet … | but Shakhan will not keep silent |
| 1084#44 | mistranslation | (copy of box 40) | Something's down here, that's for sure … | it seems certain that something is here |
| 1488#33 | omission | for whose sake he offered to undertake that difficult task | that Nikola risked his life to help escape | the daughter Nikola let escape at the risk of his life |
| 1982#17 | speaker | Hans | Captain | the Captain mocks Hans's cry |

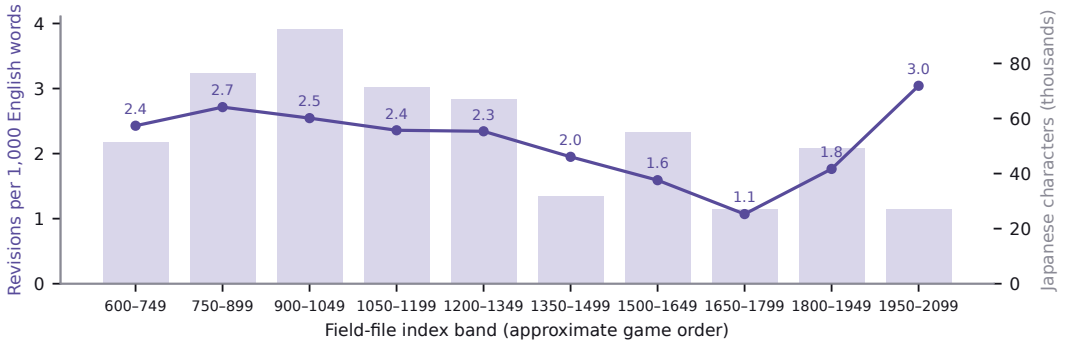



**Figure 4 | Revision density varies by a factor of three across the game.** Line: accepted revisions per 1,000 English words in each band of 150 field-file indices, with the value labelled. Bars: Japanese characters aligned in the band, in thousands. Bands are an approximation of game order. Measured data.

## 3.5. Error density is uneven along the game

We grouped files into bands of 150 field-file indices and computed accepted revisions per 1,000 English words in each band.

The overall density is 2.27 revisions per 1,000 English words, or 9.08 per 10,000 Japanese characters. The density stays between 2.34 and 2.71 in the first five bands, falls to 1.07 in the band 1650–1799, and rises to 3.04 in the last band (Figure 4).

File indices follow the order of the game only approximately, and the first band mixes pass-1 files with pass-2 files. The bands therefore describe where revisions concentrate in the script, not a measured decline in translation quality.

## 3.6. Build drift left 376 boxes of the 1998 script on disc 2

Using the rule in §2.4, we compared each disc-2 field file with its disc-1 counterpart and with both Japanese discs.

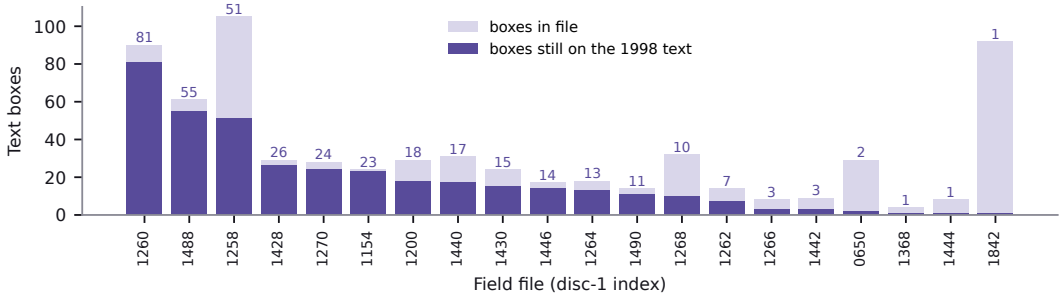



**Figure 5 | Twenty disc-2 files still shipped the 1998 script for scenes revised on disc 1.** Grey bars: text boxes in the file. Purple bars and labels: boxes whose English differed from the final disc-1 version. Files are ordered by stale-box count and named by their disc-1 index. Measured data.

Twenty disc-2 files met all three conditions. Together they hold 666 boxes, of which 376 (56.5%) still carried the 1998 localization, for example “the Ignas defeat, the Thames retreat” where disc 1 reads “the loss of Ignis …the defeat at the Tamuz” (Figure 5). Every Japanese original in these pairs is identical across the two Japanese discs. Japanese disc 2 holds the disc-1 scenes shifted by nine file indices and contains no field text that Japanese disc 1 lacks, so it adds no new source text to the audit.

In five of the twenty files the disc-1 event script also differs from the original, which is consistent with the gameplay changes the build describes [3]. Because the Japanese game uses one script for both discs, these files were copied whole, but they are the changes most in need of in-game testing.

## 4. Related work

### 4.1. Game localization

Game localization research describes the constraints of box size, control codes and release schedules that shape a game script [1]. This work adds a quantitative view of a fan retranslation, measured against the original text recovered from the game itself.

### 4.2. Translation quality frameworks

MQM organizes translation issues into a hierarchy with Accuracy and Fluency branches [7], and ISO 17100 requires revision by a second linguist who compares target and source [5]. This work applies the revision step to a complete game script and reports its outcome in categories that map onto MQM’s branches, without claiming MQM scores.

## 5. Discussion

The audit shows two things. A complete source-grounded revision of a long game script is feasible without rebuilding the script: 15,987 boxes were aligned and 498 revisions were written back as byte-level replacements. And the remaining errors in a mature

retranslation are mostly errors of meaning rather than of spelling, at a density of about two per thousand English words.

The idea that made this possible is simple: treat the disc, not a transcript, as the source of truth for both languages, and let nothing change the disc unless it applies to the exact bytes that ship. The same rule exposed the disc-2 drift, which no comparison of script files would have found.

The audit has limits. No revision has been tested in the running game. Proposals and acceptance decisions were made by SI-assisted reviewers under the author's policy, without an independent human panel, so the category counts and confidence labels carry no measured agreement. The three US-only boxes, the battle text and the five disc-2 event scripts remain open, and file-index bands are only a proxy for game order.

## 6. Methods

**Materials.** English text came from the Perfect Works Build release 0.11.4 [3], as both the shipped disc images and the project's script dump. Japanese text came from images of the two Japanese discs; disc 2 matched its Redump checksum [8]. The public transcript [6] was used only to learn the code table and to pair chapters 1–6.

**Counting rules.** A Japanese character is any decoded, non-whitespace character in a paired Japanese box, with name pointers expanded; control codes and undecoded characters are excluded. An English word is a run of letters or digits, with internal apostrophes or hyphens, after control tags are removed. Characters changed per revision is the total length of the non-matching spans returned by a sequence matcher on the tag-free old and new text.

**Verification.** After each build, every changed sector was checked to belong to an edited file or to the file table; no stray sectors were found on either disc. Error-detection and error-correction codes were regenerated and checked on sampled sectors with no mismatches. Each synced disc-2 file was confirmed byte-identical to its disc-1 counterpart.

**Statistical reporting.** All counts are censuses of the full aligned corpus, not samples, so no confidence intervals are given. Densities are ratios over the corpus or over a band. Categories and confidence labels are those assigned at proposal time. Each figure is produced by a script that reads the project's edit lists and alignment files directly.

**Use of SI.** SI systems were used to decode and align the Japanese text, to propose and vet revisions, and to build and verify the discs. The author defined the revision policy and the acceptance criteria.

## References

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